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Investor-level common ownership, stock return comovement, and competition

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Abstract
This study examines the relationship between common ownership and the correlation of stock returns. Using S&P 500 data, we find that higher levels of common ownership at the individual investor level are associated with stronger pairwise correlations of stock returns in subsequent periods. This relationship holds across multiple industries and remains robust after controlling for factors such as market risks, stock prices, liquidity, financial leverage, index additions, geographic proximity, correlated trading, and the level of competition in the industry. Our findings indicate that these effects are not solely driven by large institutional investors. Another contribution of our study is distinguishing between the effects driven by the level and the similarity of these control variables. Moreover, since reduced competition is associated with higher stock return correlation, our results are consistent with the existence of anticompetitive effects of common ownership across several industries. We provide further support for this competition channel by demonstrating that this effect is mainly driven by longer-term investors. Our findings have important policy implications, suggesting that antitrust regulations should consider indirect competitive effects from common ownership, even without explicit coordination between companies.

Keywords Anticompetitive effects · Common ownership · Pairwise correlations · Stock return comovement · Stock price synchronicity

JEL classification G23 · G28 · G32 · G38 · L41

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1 Introduction

Understanding the determinants of the correlation between stocks is crucial for portfolio allocation, as it is the decisive factor for diversification effects. In this study, we investigate the relationship between common ownership at the individual investor level and the comovement of stock returns. Using S&P 500 data from Refinitiv Eikon's Ownership Summary for the period 1998–2022, we show that increases in common ownership are followed by increases in the pairwise correlation of stock returns. This result is robust to various control variables, including Fama–French factors, stock prices, and index inclusions. Figure 1 illustrates a strong positive relationship between changes in common ownership and subsequent changes in pairwise correlation. This pattern, based on observations grouped into 100 clusters, highlights the potential relevance of ownership structure for stock return comovements and motivates our study.

Furthermore, building on the findings of Abdoh and Varela (2017), who show decreased competition is associated with higher pairwise correlations of stock returns, we link our results to the ongoing discussion on anticompetitive effects of common ownership. Specifically, our observation that increases in common ownership is followed by increases in pairwise correlation which aligns with the idea that shareholders with stakes in multiple companies use their influence on corporate policy to reduce competition between these companies to maximize the overall value of their holdings. This, in turn, leads to higher pairwise correlations of stock returns, i.e., stronger stock return comovement. In an extension to our analysis, we differentiate between short-term (holding periods up to 2 years) and non-short-term

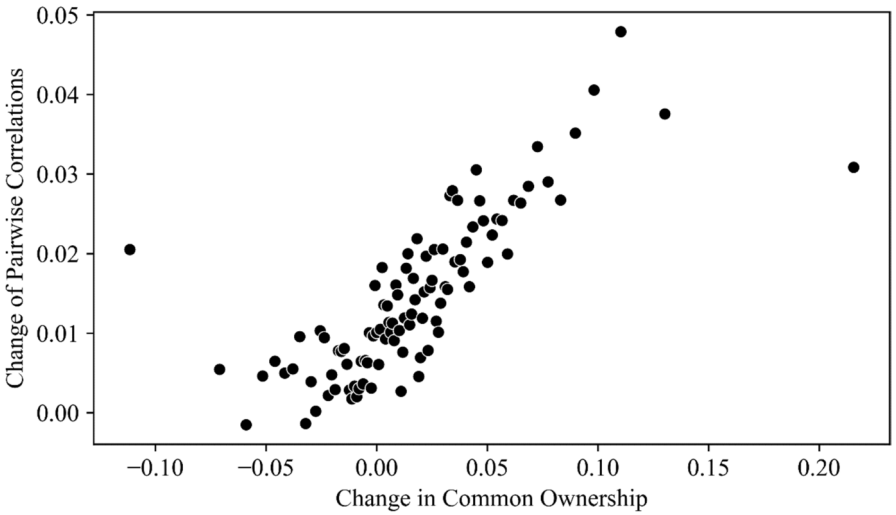

Fig. 1 Changes in pairwise correlation over changes in common ownership. This figure illustrates the positive relationship between changes in common ownership (horizontal axis) and changes in pairwise stock return correlations (vertical axis). To improve readability, observations are grouped into 100 clusters by changes in common ownership. Each point represents the mean value of both variables within a cluster

common ownership (holding periods of more than 2 years).[1] We find a significant positive effect of long-term common ownership on pairwise correlations, while the effect of short-term common ownership, depending on the specification, is negative or statistically insignificant. This result strengthens our proposed channel, as investors with longer investment horizons are more likely to exert influence on corporate policies to reduce competition.

Our key contributions are as follows. First, our analysis is not limited to specific investor types or sizes, such as actively managed or ETF funds. Instead, we provide novel analyses at the individual investor level. Notably, our results indicate that the effect of common ownership on pairwise correlations is driven by more than just a few large investors. Second, we extend the analyses of studies such as Antón and Polk (2014) by additionally considering recently developed vector-based measures for common ownership from Boller and Morton (2020) in the analysis of stock pair return correlations. Third, we use a current dataset that provides data from 1998 to 2022. This is particularly relevant given the increased relevance and influence of institutional investors in recent years. For instance, according to Lewellen and Lewellen (2022), institutional ownership of publicly traded US firms has more than doubled from 1980 to 2017. Fourth, we account for the possibility of reverse causality by considering the effect of a change in common ownership on the correlation in subsequent periods. Fifth, we include control variables that represent the pairwise similarity and the absolute level, respectively. In this way, we can show whether corporate similarity drives pairwise correlations (i.e., the similarity of stock return movements) or whether the absolute level of a particular factor is decisive. Sixth, our paper closes the gap between the analysis of stock return correlation and the discussion of anticompetitive effects of common ownership (e.g., Azar et al. 2022, 2018). Seventh, we differentiate between short-term and non-short-term common ownership, which further strengthens the proposed competition channel.

Our study is related to several streams of literature, which we present in the following paragraphs. First, we contribute to the research on stock comovement. In their seminal study, Barberis et al. (2005) present evidence on friction- or sentiment-based comovement by examining inclusions and removals of stocks in the S&P 500 index. Using data from 1979 to 2000, they show that stock returns have a higher correlation with the returns of other members of the index after the stock has been added to the S&P 500. Conversely, the correlation with the stocks that are not included in the index decreases. Green and Hwang (2009) provide evidence of price-based comovement. Specifically, they find that the stock's correlation to low-priced stocks increases after a stock split, while its correlation to stocks with higher prices decreases. This indicates that stocks within the same price category are more correlated. However, with a modified methodology, Chen et al. (2016) find limited empirical evidence for the results of Barberis et al. (2005) and Green and Hwang (2009).

Regarding the determinants of comovement, Bekaert et al. (2009) find that risk-based factor models fit stock comovements reasonably well, while studies by Chan et al. (2013) and Broman (2016) emphasize the role of liquidity. Information and

[1] We thank an anonymous reviewer and the editor, Tim Kröncke, for suggesting such an analysis.

networks also play a critical role in stock return comovement. For example, stocks affiliated with the same underwriter or with shared analyst coverage exhibit stronger comovement (Grullon et al. 2014; Muslu et al. 2014). Hameed et al. (2015) further demonstrate that firms with high analyst coverage tend to have stronger comovement with other stocks, as investors use the well-covered firms' information to value lesser-followed stocks. Other factors, such as geographical proximity (Li and Zhao 2016), commonalities in newswire text (Box 2018), and political networks (Piotroski et al. 2022), also contribute to stock return comovement.

Regarding the impact of shared ownership, Antón and Polk (2014) establish a positive relationship between shared ownership of mutual funds and the correlation of stock returns in the US stock market. In a later study, Box (2018) also finds a positive relationship between shared mutual fund and institutional ownership and stock price comovement. Ben-David et al. (2018) find that ETF membership significantly increases the volatility of the underlying securities. Analyzing data from the banking sector, Iselin et al. (2021) find that common ownership mainly via passive mutual funds increases a bank's contribution to systemic risk. Coles et al. (2022) show that index investing causes a higher correlation of returns between stocks contained in the index and with the index itself. Based on a data set of ETF holdings in the US market, Da and Shive (2018) find a positive relationship between the ETF ownership of a stock and its synchronicity with the market. They show that the comovement depends on the ETF's activity: The higher the ETF's turnover, the stronger the comovement of the shares in its portfolio. Boller and Morton (2020) create ownership vectors and use similarity measures to capture common ownership. They find that entering the S&P 500 is associated with an increase in common ownership and stock returns. Furthermore, they show that their measures of common ownership predict cumulative abnormal returns.

The aforementioned studies each focus on a specific type of common ownership, such as through mutual funds or ETFs. In contrast, we are among the first to consider the overall stock holdings of investors, regardless of the investment type. This approach provides novel evidence on the effects of common ownership at the individual investor level. Considering common ownership at the investor level is crucial because it enables us to link it to anticompetitive effects through investors' influence on corporate policies. Another distinction from previous studies is that, due to this mechanism, we do not weight common ownership by the market values of the two companies. Weighting by market values causes common ownership to be driven by the share in the larger company, which fails to account for the fact that influencing the policy of the smaller company can significantly impact the competition and, consequently, the correlation between both companies. Our approach captures the possibility of such anticompetitive effects more effectively.

Second, our study is related to the several articles that study stock price synchronicity at the market level. Based on the approach of Roll (1988), Morck et al. (2000) and Jin and Myers (2006) empirically investigate stock price synchronicity using the R-squared from cross-sectional regressions. Jin and Myers (2006) argue that the less transparent the company is, the more of its operating cash flow can be captured by internal managers. Thus, less firm-specific risk is absorbed by external investors. This leads to the opacity of companies being linked to stock price synchronicity. An

and Zhang (2013) extend this work and establish a connection to institutional investors by using the classification of Bushee (1998) for dedicated and transient institutional investor as well as quasi-indexers. While dedicated institutional investors monitor the extraction of cash flows, transient institutional investors trade frequently and have less incentive to monitor the companies they have invested in. Analyzing data from 1987 to 2010, An and Zhang (2013) find a negative relationship between dedicated institutional investors and stock price synchronicity. However, the presence of transient institutional investors is accompanied by an increase in stock price synchronicity. In contrast with the presented research, our study does not analyze correlations at the market level but instead focuses on pairwise correlations, i.e., correlations at the company level.

Third, our study also contributes to the literature on anticompetitive effects of common ownership. Azar et al. (2022) and Azar et al. (2018) analyze the impact of common ownership on competition in the banks and airline industry. In both industries, an increase in common ownership results in higher prices, likely via lower product market competition. In contrast, other studies, such as Dennis et al. (2022), Gramlich and Grundl (2018), and Backus et al. (2021) provide limited empirical support for anticompetitive effects of common ownership in the airline, banking, and ready-to-eat industry, respectively.

Regarding the link between pairwise correlation and competition, Abdoh and Varela (2017) analyze data from 2005 to 2014. They show that increasing product market competition measured with the Herfindahl–Hirschman index (HHI) is associated with an increase in idiosyncratic volatility relative to systematic volatility, i.e., a decrease in pairwise correlation. A possible explanation is that when market power and concentration are high, firm-specific cost shocks can be more easily passed on to customers because fewer substitutes are available to them. When competition and pairwise correlations have a negative relationship, and common ownership leads to a lower level of competition, pairwise correlations should be positively related to common ownership. Therefore, our analysis can be interpreted as evidence on anticompetitive effects of common ownership. Note that our analysis includes changes in industry HHIs as control variables, meaning that our results do not merely reflect effects of industry-level competition.

The remainder of this study is organized as follows. Section 2 describes the data set. Section 3 presents the empirical method. In Sect. 4, the results are described. Section 5 contains a discussion, and Sect. 6 concludes.

2 Data set

We use data from the US stock market. Specifically, we consider stocks listed in the S&P 500 index. In addition to reducing the high effort required for data collection and processing, focusing on the S&P 500 has the following advantages. First, the index contains stocks from multiple industries, and covers the industries of airlines and banks examined by Azar et al. (2018) and Azar et al. (2022). Second, the index is a good single indicator of large capitalization equities, covering approximately

80% of the available market capitalization in the USA.[2] Therefore, our results are likely to reflect the behavior of large investors and institutional investors, which means that they are likely to apply to other indices, stock markets, and countries. Third, the data regarding shareholder reports on the investor level for stocks contained in the S&P index are reliably available. Specifically, for each stock, quarterly ownership reports can be collected from Refinitiv Eikon's[3] Ownership Summary ("Shareholder History Report"). Each report contains all individual shareholders of a company with their name and the percentage of outstanding shares held. In contrast with the Thomson Reuters Spectrum data set, which has been used in several related studies (e.g., Antón and Polk 2014; Azar et al. 2018), the Ownership Summary is not limited to 13F filings on institutional investors, but also includes insider trading, mutual funds, and private investors. This provides a more holistic view of changes in shareholder composition and allows us to investigate common ownership at the individual investor level.

In order to account for different notations, all investor names are converted to lower case, and special characters are removed. However, for complex investor networks, e.g., due to globally operating companies with national subsidiaries, the corresponding investor names vary. Thus, the investor names are manually harmonized by replacing different tightly connected investor names with the corresponding umbrella organization. For example, various Blackrock subsidiaries such as "BlackRock Investment Management (UK) Ltd." and "BlackRock Investment Management (Australia) Ltd." are replaced with "blackrock" as the investor name. However, our results are robust to using the raw investor names. The corresponding results are presented in Appendix A. After the adjustments, the ownership share is calculated as the sum over each unified investor name.

To calculate stock return comovement, the time series of daily closing prices "P," which are adjusted for capital measures, are retrieved and used to calculate stock returns. The return $R_{i,t}$ for stock i at time t is calculated according to the following equation[4]:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}. \tag{1}$$

The analysis focuses on competing companies with similar business areas, i.e., companies with similar products and services. Thus, industry classifications are retrieved. In particular, the 4-digit ICB super-sector codes are collected for all relevant stocks. This choice addresses the following trade-off: On the one hand, the more granular the ICB code, the more similar are the business areas of two companies, and thus, the greater is the probability that the two companies are competitors. On the other hand, the more specific the ICB code, the smaller is the number of stocks that are grouped in this ICB classification. Due to the limited

[2] See https://www.spglobal.com/spdji/en/indices/equity/sp-500/#overview, last accessed August 6, 2024.

[3] Note that Refinitiv Eikon was previously known as Thomson Reuters Eikon.

[4] Note that, in order to avoid biased results because of outliers, a stock's calculated return time series is truncated at its 1%- and 99%-percentile.

Table 1 Industry split of companies for HHI calculation

Industry super-sector	Number of companies
1010—Technology	769
1510—Telecommunications	201
2010—Health Care	1273
3010—Banks	620
3020—Financial services	1429
3030—Insurance	122
3510—Real Estate	331
4010—Automobiles and parts	77
4020—Consumer products and services	398
4030—Media	149
4040—Retail	195
4050—Travel and leisure	232
4510—Food, beverage and tobacco	244
4520—Personal care, drug and grocery stores	85
5010—Construction and materials	154
5020—Industrial goods and services	854
5510—Basic resources	226
5520—Chemicals	91
6010—Energy	537
6510—Utilities	168

amount of available data, this may result in an insufficient number of observations per classification code to perform an empirical analysis.

We collect the daily market value "*MV*," price to book value "*PTBV,*" operating profitability margin "*OPM,*" total assets "*TA,*" leverage ratio "*LR,*" the number of shares traded "*VO,*" the history of additions "*Add*" to the S&P 500 index, and the US state of a firm's corporate office "*LOC(State)"* for each stock as control variables. The time series are truncated at the 0.5%- and 99.5%-quantile.

The period under consideration extends from 1998 to 2022. We use Refinitiv Eikon's *Datastream* to retrieve data on the presented variables of all stocks included in the S&P 500. Descriptive statistics for the data set will be presented after we introduce all computed variables in the next section. Furthermore, to calculate the Herfindahl–Hirschman Index (HHI) for each industry as a control for competition, we collect quarterly market values of all companies with headquarters in the USA in the respective ICB super-sectors that are available on Eikon's *Datastream*. Overall, we consider more than 8,000 companies. The split of these companies across the super-sectors is presented in Table 1.

3 Methodology

This section presents our methodology to analyze the hypothesized relationship between common ownership at the investor level and stock return comovement. Instead of focusing on specific events, such as the addition to an index, we consider changes in common ownership and correlation over a period of 24 years. Our methodology consists of three parts. First, all relevant variables are computed for each time point. Second, the change of these variables over a specified period is calculated. Third, the relationship between the change of common ownership of a stock pair and the subsequent change of pairwise return correlations is estimated by means of linear regression.

We create a subsample for each industry classification. Within each industry classification, all possible combinations of stocks are pooled to create stock pairs. We then calculate all variables for these stock pairs. We measure common ownership in three ways. First, we consider ownership overlap by adopting the formula of Münster and Walther (2021)[5]:

$$SHO_{i,j}^{t} = \sum_{k} \left(1 - \frac{\delta_{k,i,j}^{t}}{S_{i,k}^{t} + S_{j,k}^{t}} \right) \cdot \frac{S_{i,k}^{t} + S_{j,k}^{t}}{2} = \min(S_{i,k}^{t}, S_{j,k}^{t}) \forall i, j, t. \tag{2}$$

$S_{i,k}^{t}$ represents the percentage of stocks of company i held by investor k at time t. We assume that investors' influence on a pair of companies can be exerted mainly when they hold a significant stake in both stocks. Since an investor's average holdings $\frac{S_{i,k}^{t}+S_{j,k}^{t}}{2}$ can be biased when only one of the stakes is large, the measure above also takes the difference between the stakes $\delta_{k,i,j}^{t} = \left| S_{i,k}^{t} - S_{j,k}^{t} \right|$ into account. By rescaling with the factor $\left(1 - \frac{\delta_{k,i,j}^{t}}{S_{i,k}^{t}+S_{j,k}^{t}} \right)$, the measure counteracts the potential bias because the value of the factor increases (decreases) when the difference between the two stakes is small (large). SHO corresponds to the sum across all investors of an investor's minimum percentages of stocks in both companies. It therefore intuitively reflects the overlap between the owners of both companies. However, the minimum does not take asymmetric holdings into account. These might be relevant for anti-competitive effects as holding a large stake in one company might already enable an investor to influence the competitive strategy of the company and thus the correlation between both stocks. Therefore, in Appendix B, we provide two robustness checks. First, we measure common ownership as the sum across all investors of the average percentages of stocks in both companies. This means that we drop the penalty term. Second, we consider a measure that uses half the penalty term, i.e., $\left(1 - \frac{1}{2} \cdot \frac{\delta_{k,i,j}^{t}}{S_{i,k}^{t}+S_{j,k}^{t}} \right)$. It corresponds to the average of the two former measures and rep-

[5] See also Antón and Polk (2014) for a related definition. However, the authors make use of a weighting of shares with market values.

resents a compromise between both approaches.[6] All three variants yield similar results.

In contrast with Antón and Polk (2014), SHO does not assign a larger weight to larger companies, which means that SHO is not mainly driven by the stake in the larger company. Moreover, SHO ensures that investors hold a sufficient stake in both companies so that there may be an interest in collusive behavior in the first place. This is not the case with the measure of Antón and Polk (2014) as the distribution of ownership across both companies is irrelevant (Gilje et al. 2018).

The second and third measures are based on ownership similarity. We employ these vector-based measures of common ownership, following the methodology of Boller and Morton (2020), for two reasons. First, these measures compare the entire ownership structures of companies rather than focusing solely on investor overlap. Second, as shown by Boller and Morton (2020), only vector similarity measures exhibit a significant correlation with abnormal returns, emphasizing their relevance for analyzing potential anticompetitive effects.

The measures are calculated as follows. For both, all investors are encoded in an ownership vector. For computational reasons, only the most frequent 5000 investors from the entire data set are considered, resulting in an ownership vector with $K = 5{,}000$ dimensions. Note that the results are robust to choosing different vector sizes. The corresponding results for $K = 1{,}500$ and $K = 15{,}000$ are presented in Appendix C. Each dimension of the vector is assigned an investor and the share of outstanding shares. Thus, the ownership vector of stock i at time point t can be written as:

$$\overline{OS}_i = \begin{pmatrix} S^t_{i,k=1} \\ \dots \\ S^t_{i,k=K} \end{pmatrix} \forall i, t. \tag{3}$$

Using the ownership vectors defined above, we calculate the second and third common ownership measures following Boller and Morton (2020). Both are similarity-based but capture similarity through different methods. The second measure is derived from the comparison of the vectors based on cosine similarity:

$$\mathrm{SHC}^t_{i,j} = \frac{\overline{OS_{i,t}} \cdot \overline{OS_{j,t}}}{\|\overline{OS_{i,t}}\| \|\overline{OS_{j,t}}\|} = \frac{\sum_k S^t_{i,k} \cdot S^t_{j,k}}{\sqrt{\sum_k \left(S^t_{i,k}\right)^2} \sqrt{\sum_k \left(S^t_{j,k}\right)^2}} \forall i, j, t. \tag{4}$$

Cosine similarity is more affected by deviations of large investors than of small investors. In contrast, the Bray–Curtis distance, first described by Bray and Curtis (1957), weights all deviations equally, regardless of the investment amount. Thus, the last variable is derived from a comparison of the vectors by means of the Bray–Curtis similarity:

[6] For a discussion of pros and cons of a common ownership measure based on the minimum ownership in both companies see Gilje et al. (2018).

$$SHB_{i,j}^{t} = 1 - \frac{\sum_k \left| S_{i,k}^{t} - S_{j,k}^{t} \right|}{\sum_k \left| S_{i,k}^{t} + S_{j,k}^{t} \right|} \forall i, j, t. \tag{5}$$

Both cosine and Bray–Curtis similarity can range from zero to one. A similarity of zero indicates that the companies have no common owners, implying no common ownership, while a similarity of one implies identical shareholder structures. Therefore, a high similarity score suggests that a large proportion of investors may have an incentive to avoid competition between the companies. Conversely, a low similarity score indicates that no such incentives are likely to exist.

All three measures of common ownership are computed for all stock pairs grouped according to their industry classification. Furthermore, we compute Pearson correlation coefficients based on daily returns as the dependent variable:

$$\mathrm{CORR}_{i,j} = \frac{Cov\left(R_i, R_j\right)}{\sqrt{Var(R_i)} \cdot \sqrt{Var(R_j)}} \forall i, j. \tag{6}$$

In the following, we present and explain the used control variables. Common factors (Fama and French 1993) are likely to have an effect on pairwise correlations. To ensure that these effects are not the drivers of our results on common ownership, we include control variables for factor similarity into our regressions.[7] Specifically, we control for the common factors presented in Fama and French (2015). As Antón and Polk (2014), we compute percentiles for each control variable and period to eliminate the impact of outliers. Furthermore, for each factor, we consider the similarity between both companies (the absolute difference in the respective factor), and the absolute level, respectively. In this way, we can test whether similarity in fundamentals drives correlations (i.e., the similarity of stock return movements), or whether particular levels of the respective factor increase comovement. We expect a positive relationship between correlation and the similarity of factors, while the effect of the factor level depends on the specific variable.

The first considered factor represents a company's market value, $\mathrm{MV}_{i,t}$.We define the corresponding regression variable as the absolute value of the distance between the market value percentiles ($\mathrm{MV}_{\mathrm{PERCENTILE}i,t}$) for each stock pair

$$\mathrm{MV}_{\mathrm{DIS}}{}_{i,j}^{t} = \left| \mathrm{MV}_{\mathrm{PERCENTILE}i,t} - \mathrm{MV}_{\mathrm{PERCENTILE}j,t} \right| \forall i, j, t. \tag{7}$$

Following Green and Hwang (2009), we additionally control for absolute market values by computing the average market value percentile $\mathrm{MV}_{\mathrm{AVG}}{}_{i,j}^{t}$ for each stock pair

$$\mathrm{MV}_{\mathrm{AVG}}{}_{i,j}^{t} = \frac{1}{2} \cdot \left(\mathrm{MV}_{\mathrm{PERCENTILE}i,t} + \mathrm{MV}_{\mathrm{PERCENTILE}j,t} \right) \forall i, j, t. \tag{8}$$

[7] See also Antón and Polk (2014) for a similar approach.

The intuition behind using the average rather than the minimum is that even if only one of the two companies is large and thus likely to operate in several different lines of business, the likelihood of overlapping businesses with the other company increases.

Similarly, we control for the price to book value $\mathrm{PTBV}_{i,j}^{t}$ by including the following control variables

$$\mathrm{PTBV}_{\mathrm{DIS}_{i,j}^{t}} = \left|\mathrm{PTBV}_{\mathrm{PERCENTILE}_{i,t}} - \mathrm{PTBV}_{\mathrm{PERCENTILE}_{j,t}}\right| \forall i, j, t \tag{9}$$

$$\mathrm{PTBV}_{\mathrm{AVG}_{i,j}^{t}} = \frac{1}{2} \cdot \left(\mathrm{PTBV}_{\mathrm{PERCENTILE}_{i,t}} + \mathrm{PTBV}_{\mathrm{PERCENTILE}_{j,t}}\right) \forall i, j, t. \tag{10}$$

The control variables on the operating profitability margin $\mathrm{OPM}_{i,j}^{t}$ (profitability) are defined as

$$\mathrm{PTBV}_{\mathrm{DIS}_{i,j}^{t}} = \left|\mathrm{PTBV}_{\mathrm{PERCENTILE}_{i,t}} - \mathrm{PTBV}_{\mathrm{PERCENTILE}_{j,t}}\right| \forall i, j, t, \tag{11}$$

$$\mathrm{PTBV}_{\mathrm{AVG}_{i,j}^{t}} = \frac{1}{2} \cdot \left(\mathrm{PTBV}_{\mathrm{PERCENTILE}_{i,t}} + \mathrm{PTBV}_{\mathrm{PERCENTILE}_{j,t}}\right) \forall i, j, t, \tag{12}$$

while investment is proxied by the change of total assets $TA_{i,j}^{t}$

$$\mathrm{TA}_{\mathrm{DIS}_{i,j}^{t}} = \left|\mathrm{TA}_{\mathrm{PERCENTILE}_{i,t}} - \mathrm{TA}_{\mathrm{PERCENTILE}_{j,t}}\right| \forall i, j, t, \tag{13}$$

$$\mathrm{TA}_{\mathrm{AVG}_{i,j}^{t}} = \frac{1}{2} \cdot \left(\mathrm{TA}_{\mathrm{PERCENTILE}_{i,t}} + \mathrm{TA}_{\mathrm{PERCENTILE}_{j,t}}\right) \forall i, j, t. \tag{14}$$

In addition to common factors, we control for different price levels since Green and Hwang (2009) find a connection between stock price similarity and comovement. For this purpose, we calculate the absolute stock price difference $PD_DIS^{t}{}_{i,j}$ between stock prices ($P_PERCENTILE_{i,t}$) for each firm pair. Analogous to the other control variables, we also calculate the mean price difference:

$$P^{t}_{\mathrm{DIS}_{i,j}} = \left|P_{\mathrm{PERCENTILE}_{i,t}} - P_{\mathrm{PERCENTILE}_{j,t}}\right| \forall i, j, t \tag{15}$$

$$P^{t}_{\mathrm{AVG}_{i,j}} = \frac{1}{2} \cdot \left(P_{\mathrm{PERCENTILE}_{i,t}} + P_{\mathrm{PERCENTILE}_{j,t}}\right) \forall i, j, t. \tag{16}$$

As control variables for liquidity, we add trading volume (VO) to the regressions. This helps mitigate potential confounding effects on common ownership, such as common investors preferring stocks with similar liquidity levels due to trading constraints or investment strategies:

$$VO_{\mathrm{DIS}_{i,j}} = \left|VO_{\mathrm{PERCENTILE}_{i,t}} - VO_{\mathrm{PERCENTILE}_{j,t}}\right| \forall i, j, t \tag{17}$$

$$\mathrm{VO}^t_{\mathrm{AVG}i,j} = \frac{1}{2} \cdot \left(VO_{\mathrm{PERCENTILE}i,t} + VO_{\mathrm{PERCENTILE}j,t} \right) \forall i, j, t. \tag{18}$$

To address the possibility that correlated trading is the cause of changes in stock return comovement, we additionally control for the correlation of the traded volume (VO) of a stock pair:

$$\mathrm{CORR_}VO^t{}_{i,j} = \frac{Cov\left(\mathrm{VO}_i, \mathrm{VO}_j\right)}{\sqrt{Var(\mathrm{VO}_i)} \cdot \sqrt{Var(\mathrm{VO}_j)}} \forall i, j, t. \tag{19}$$

The leverage ratio of a company (LR_i) measures the proportion of debt capital to total capital and allows us to control for companies' financial risks. Including leverage helps mitigate potential confounding effects on common ownership, such as common investors favoring firms with similar leverage levels due to risk preferences. We include both the distance and the mean value of the leverage ratios of a stock pair in our regressions:

$$\mathrm{LR}^t_{\mathrm{DIS}i,j} = \left| \mathrm{LR}_{\mathrm{PERCENTILE}i,t} - \mathrm{LR}_{\mathrm{PERCENTILE}j,t} \right| \forall i, j, t, \tag{20}$$

$$LR^t_{\mathrm{AVG}i,j} = \frac{1}{2} \cdot \left(\mathrm{LR}_{\mathrm{PERCENTILE}i,t} + \mathrm{LR}_{\mathrm{PERCENTILE}j,t} \right) \forall i, j, t. \tag{21}$$

To control for the level of competition per industry, we calculate the Herfindahl–Hirschman index (HHI) per industry:

$$\mathrm{HHI}^t_{\mathrm{Ind}} = \sum_{i \in Ind} W^2_{i,t} \forall Ind, t, \tag{22}$$

where $W_{i,t}$ denotes the market share percentage, based on market values, of company i in period t, and Ind represents the set of companies in the respective industry. To avoid skewness in the dependent variables, the HHI values are normalized by dividing by the maximum value observed in each period.

In the second step of our procedure, the variables defined in the previous step are compared over time. For this purpose, we compare ownership structure and stock return comovement over a moving time window of one year. The intuition behind this is as follows: We assume that a potential change in corporate strategy and behavior, e.g., a change in pricing or product strategy due to a change in common ownership, requires a longer period of time to be implemented in a company's operations and to show an observable effect in the market. Note that the results remain robust across different timeframes. For instance, using a time window of two years yields qualitatively comparable results, although the coefficients are slightly smaller. Detailed tables are provided in Appendix D.

Eikon provides ownership data with quarterly granularity. Our last complete quarter ends in June 30 2022. Therefore, we use this date as boundary date, as this maximizes the number of useable time frames. Consequently, the first timeframe compares 1998-06-30 and 1999-06-30, and the second timeframe compares

1999-06-30 with 2000-06-30, up to the last timeframe comparing 2021-06-30 and 2022-06-30.

In contrast with Antón and Polk (2014), our analysis focuses on the change of variables instead of their absolute values. First, the change in common ownership between two points in time is calculated. Second, the subsequent change of the pairwise stock return correlation is computed. In the last step, we analyze whether a change in common ownership is followed by a variation in the pairwise stock return correlation in future periods. Our approach addresses the possibility of reverse causality, without having to rely on external events, such as the addition to an index. The intuition behind it is as follows. If we observe that a change in common ownership is followed by a change in pairwise correlations, it is more likely that the change in common ownership led to the subsequent change in correlation than that investors deliberately chose to invest in stocks with a higher correlation in future periods. Another argument against reverse causality is that risk-averse investors have an incentive to invest in stocks with low correlation in order to minimize the variance of portfolio returns, which means that from a portfolio theory perspective, one would expect a negative relationship between common ownership and correlation (see, e.g., Münster and Walther 2021).

To capture the change in common ownership between the two time points t and $t-1$, the difference is computed as follows:

$$\Delta\mathrm{SHO}_{i,j}^{t_b} = \mathrm{SHO}_{i,j}^{t_b} - \mathrm{SHC}_{i,j}^{t_b-1} \forall i, j, t_b \in T_b,$$

$$\Delta\mathrm{SHC}_{i,j}^{t_b} = \mathrm{SHC}_{i,j}^{t_b} - \mathrm{SHC}_{i,j}^{t_b-1} \forall i, j, t_b \in T_b,$$

$$\Delta\mathrm{SHB}_{i,j}^{t_b} = \mathrm{SHB}_{i,j}^{t_b} - \mathrm{SHB}_{i,j}^{t_b-1} \forall i, j, t_b \in T_b. \tag{23}$$

Analogously, the changes of the control variables for the market value ($\Delta\mathrm{MV}_{\mathrm{DIS}_{i,j}}^{t_b}$, $\Delta\mathrm{MV}_{\mathrm{AVG}_{i,j}}^{t_b}$) are computed for each stock pair and timeframe as the difference between the respective values at the boundary dates:

$$\Delta\mathrm{MV}_{DIS_{i,j}}^{t_b} = \mathrm{MV}_{\mathrm{DIS}_{i,j}}^{t_b} - \mathrm{MV}_{\mathrm{DIS}_{i,j}}^{t_b-1} \forall \mathrm{i, j}, t_b \in \mathrm{T_b},$$

$$\Delta\mathrm{MV}_{\mathrm{AVG}_{i,j}}^{t_b} = \mathrm{MV}_{\mathrm{AVG}_{i,j}}^{t_b} - \mathrm{MV}_{\mathrm{AVG}_{i,j}}^{t_b-1} \forall i, j, t_b \in T_b, \tag{24}$$

Note that due to days without trading, such as bank holidays, data for the market value may be missing on some boundary dates. Therefore, the percentile ranking for the market value $\mathrm{MV_PERCENTILE}_i^{t_b}$ is calculated as the average value considering a timeframe including the surrounding six days:

$$MV_{\mathrm{PERCENTILE}i}^{t_b} = \frac{1}{7} \sum_{t=t_b-3}^{T=t_b+3} MV_{\mathrm{PERCENTILE}i}^{t_b}, \tag{25}$$

Analogously, all other control factors, with the exception of the correlation of trading volumes, are preprocessed and converted into delta variables as described above.

Barberis et al. (2005) show that the addition to an index affects the comovement of the underlying stocks. Furthermore, Boller and Morton (2020) demonstrate that the addition to the S&P 500 is accompanied by an increase in common ownership. Thus, we control for the addition to the S&P 500. The dummy variable $Add_i^{t_b-1,t_b}$ has a value of one when the stock i has been added to the S&P 500 between $t_b - 1$ and t_b; otherwise, it is zero. To account for the pair of stocks, the control variable considers whether any of the two stocks has been added to the index:

$$\mathrm{ADD}_{i,j}^{t_b-1,t_b} = \begin{cases} 1, \text{if } Add_i^{t_b-1,t_b} + Add_j^{t_b-1,t_b} \geq 1 \\ 0, else \end{cases}. \tag{26}$$

Li and Zhao (2016) demonstrate the relevance of geographic proximity of companies for stock price correlations. We account for this by adding a dummy variable that captures if the corporate offices of the two companies in a pair are located in the same state in the USA:

$$\mathrm{SAME_LOC}_{i,j}^{t_b} = \begin{cases} 1, \text{if } LOC(State_i) = LOC(State_j) \\ 0, else \end{cases}. \tag{27}$$

The dependent variable of the analysis is the pairwise stock return correlation, which cannot be calculated exactly for a specific time point but instead needs to be estimated with a time series. Therefore, the pairwise stock return correlation for a boundary date is estimated with the subsequent year of daily returns. Thus, for each boundary date, a new timeframe T_b is created by adding one year to the boundary date[8]:

$$T_b = \left[t_b \ldots t_b + 1year\right]. \tag{28}$$

The pairwise correlation for a boundary date and stock pair $\mathrm{CORR}_{i,j}{}^{t_b}$ is estimated according to Eq. (6) based on the return time series including the subsequent year of daily returns of the two stocks $R_{i,t}, R_{j,t} \forall t \in T_b$. The idea of considering subsequent return data is that our analysis focuses on changes in correlations following changes in common ownership. However, as a robustness check, we also estimate the correlation using a time window surrounding the boundary date. The qualitative results remain unchanged and are provided in Appendix E. When using a surrounding time window, the timeframe T_b is created by adding or subtracting half a year to the boundary date, respectively:

$$T_b = \left[t_b - 0.5year \ldots t_b + 0.5year\right]. \tag{29}$$

[8] The only exception is the last boundary date 2022–30-06. Here, all returns available future returns until 2022–10-31 are used.

Analogously to the independent variables, the change in the pairwise stock return correlation, our dependent variable, is computed as

$$\Delta\text{CORR}_{i,j}^{t_b} = \text{CORR}_{i,j}^{t_b} - \text{CORR}_{i,j}^{t_b-1}. \tag{30}$$

Using the same logic, we introduce an additional control variable that captures correlated trading volume as a possible determinant of correlated stock returns. It is constructed by first computing the pairwise correlation between trading volumes. To have consistent control variables, we convert these correlations to percentiles and then calculate the change in these percentiles:

$$\Delta CORR_{\text{VO}}^{t_b} = PERCENTILE\left(\text{CORR}_{\text{PERCENTILE}_{VO_{i,j}}}^{t_b}\right)$$

$$-PERCENTILE\left(\text{CORR}_{\text{PERCENTILE}_{VO_{i,j}}}^{t_b-1}\right). \tag{31}$$

Finally, a data point of the data set is described by its date boundaries $(t_b, t_b - 1year)$, three measures for the variation in common ownership $\left(\Delta\text{SHO}_{i,j}^{t_b}, \Delta SHC_{i,j}^{t_b}, \Delta SHB_{i,j}^{t_b}\right)$, the delta variables computed analogously to Eq. (24) for the control factors defined in Eqs. (7) to (21), the indicators for index additions $ADD_{i,j}^{t_b-1,t_b}$ and geographic proximity $\text{SAME_LOC}_{i,j}^{t_b}$, the change in the correlation of traded volume $\Delta\text{CORR}_{\text{VO}}^{t_b}$, and the change in the pairwise stock return correlation $\Delta\text{CORR}_{i,j}{}^{t_b}$. A data point is created for each industry classification, timeframe, and stock pair. Tables 2 and 3 present descriptive statistics and the correlation matrix.

In the third step of the methodology, regressions are performed to investigate the relationship between common ownership and stock return comovement. Initially, we use separate univariate regressions with each measure of common ownership, i.e., $\Delta\text{SHM} = \Delta\text{SHO}, \Delta\text{SHC} or \Delta\text{SCB}$, respectively. In Eq. (32), a_{OS} denotes the intercept and b_{OS} denotes the slope of the common ownership measure. Error terms are denoted by ε.

$$\Delta\text{CORR}_{i,j}^{t_b} = a_{\text{OS}} + b_{\text{OS}} \cdot \Delta SHM_{i,j}^{t_b} + \varepsilon_{i,j}^{t_b}. \tag{32}$$

We control for period-dependent variations in comovement, such as those arising from macroeconomic shocks (e.g., higher market volatility during crises, as discussed by Forbes and Rigobon 2002) or shifts in investor sentiment by including time fixed effects, i.e., dummies for each time period, λ_{t_b}. Note that we remove the intercept to avoid multicollinearity:

$$\Delta\text{CORR}_{i,j}^{t_b} = b_{\text{OS}} \cdot \Delta SHM_{i,j}^{t_b} + \lambda_{t_b} + \varepsilon_{i,j}^{t_b}. \tag{33}$$

As an additional time-related robustness check, we use the approach of Antón and Polk (2014), which was originally developed by Fama and MacBeth (1973). In particular, we run the regressions for each period separately and report the average

Table 2 Descriptive statistics

No	Parameter	Mean	Standard Deviation	Min	Max	10th Quantile	90th Quantile
	SHO	0.355	0.121	0.000	0.931	0.200	0.494
	SHC	0.581	0.210	0.000	1.000	0.278	0.842
	SHB	0.453	0.129	0.000	0.897	0.284	0.609
	P_{DIS}	0.295	0.219	0.000	0.990	0.042	0.620
	P_{AVG}	0.570	0.212	0.010	1.000	0.275	0.849
	$\mathrm{MV}_{\mathrm{DIS}}$	0.313	0.231	0.000	0.990	0.044	0.656
	$\mathrm{MV}_{\mathrm{AVG}}$	0.571	0.206	0.010	1.000	0.288	0.847
	$\mathrm{PTBV}_{\mathrm{DIS}}$	0.278	0.218	0.000	0.990	0.038	0.602
	$\mathrm{PTBV}_{\mathrm{AVG}}$	0.564	0.210	0.010	1.000	0.265	0.832
	$\mathrm{PTBV}_{\mathrm{DIS}}$	0.290	0.222	0.000	0.990	0.040	0.622
	$\mathrm{PTBV}_{\mathrm{AVG}}$	0.536	0.201	0.010	1.000	0.270	0.810
	$\mathrm{LR}_{\mathrm{DIS}}$	0.285	0.228	0.000	0.990	0.035	0.622
	$\mathrm{LR}_{\mathrm{AVG}}$	0.479	0.225	0.010	1.000	0.180	0.791
	$\mathrm{TA}_{\mathrm{DIS}}$	0.305	0.236	0.000	0.990	0.040	0.670
	$\mathrm{TA}_{\mathrm{AVG}}$	0.515	0.221	0.010	1.000	0.210	0.820
	VO_{DIS}	0.305	0.225	0.000	0.984	0.044	0.638
	VO_{AVG}	0.497	0.215	0.012	1.000	0.211	0.797
	HHI	0.227	0.154	0.040	1.000	0.077	0.373
	$\mathrm{CORR}_{\mathrm{VO}}$	0.545	0.287	0.000	1.000	0.131	0.929
	CORR	0.459	0.187	−0.432	0.995	0.219	0.709
1	ΔSHO	0.016	0.044	−0.375	0.553	−0.029	0.063
2	ΔSHC	0.024	0.102	−0.971	0.800	−0.089	0.137
3	ΔSHB	0.016	0.051	−0.411	0.663	−0.038	0.072
4	ΔP_{DIS}	−0.004	0.119	−0.846	0.788	−0.140	0.134
5	ΔP_{AVG}	0.010	0.076	−0.649	0.617	−0.076	0.098
6	$\Delta\mathrm{MV}_{\mathrm{DIS}}$	−0.005	0.093	−0.872	0.716	−0.112	0.100
7	$\Delta\mathrm{MV}_{\mathrm{AVG}}$	0.009	0.056	−0.527	0.553	−0.054	0.074
8	$\Delta\mathrm{PTBV}_{\mathrm{DIS}}$	−0.000	0.155	−0.990	0.990	−0.158	0.156
9	$\Delta\mathrm{PTBV}_{\mathrm{AVG}}$	0.002	0.110	−0.990	0.980	−0.098	0.102
10	$\Delta\mathrm{PTBV}_{\mathrm{DIS}}$	−0.001	0.144	−0.930	0.900	−0.150	0.146
11	$\Delta\mathrm{PTBV}_{\mathrm{AVG}}$	0.002	0.094	−0.875	0.768	−0.090	0.095
12	$\Delta\mathrm{LR}_{\mathrm{DIS}}$	0.000	0.155	−0.989	0.990	−0.146	0.149
13	$\Delta\mathrm{LR}_{\mathrm{AVG}}$	0.002	0.106	−0.980	0.949	−0.082	0.097
14	$\Delta\mathrm{TA}_{\mathrm{DIS}}$	−0.001	0.057	−0.752	0.772	−0.050	0.044
15	$\Delta\mathrm{TA}_{\mathrm{AVG}}$	0.003	0.031	−0.187	0.441	−0.024	0.030
16	ΔVO_{DIS}	−0.001	0.132	−0.822	0.874	−0.162	0.158
17	$\Delta\mathrm{VO}_{\mathrm{AVG}}$	−0.001	0.078	−0.437	0.616	−0.094	0.093
18	ΔHHI	−0.008	0.101	−0.499	0.677	−0.082	0.054
19	$\Delta\mathrm{CORR}_{\mathrm{VO}}$	0.005	0.314	−1.000	1.000	−0.404	0.424
20	SAME_*LOC*	0.100	0.301	0.000	1.000	0.000	1.000
21	ADD	0.056	0.229	0.000	1.000	0.000	0.000

Table 2 (continued)

No	Parameter	Mean	Standard Deviation	Min	Max	10th Quantile	90th Quantile
22	$\Delta CORR$	0.015	0.160	−0.804	0.634	−0.205	0.208

The descriptive statistics for each variable are computed with all $n = 146,052$ data points in our sample. The measures of common ownership SHO, *SHC*, and SHB are defined according to Eqs. (2), (4), and (5), respectively. The corresponding changes in common ownership ΔSHO, Δ*SHC*, and ΔSHB are calculated according to Eq. (23). All other control variables are calculated according to Eqs. (7) to (21), (26), (27), and (30). The pairwise changes of the control variables are computed according to Eqs. (24) and (32). Analogously, the Pearson correlation coefficient for pairwise stock returns (Corr) and its variation ΔCorr are computed as described in Eqs. (6) and (30). The first column assigns variables to numbers that are used in the correlation matrix

of the estimated coefficients. To evaluate the statistical significance of the resulting coefficients, a two-sided t test is performed using the time series of estimates. As Antón and Polk (2014), we calculate standard errors according to Newey and West (1987).

In addition to the time fixed effects, we include the presented control variables, i.e., the common risk factors of Fama and French (2015), stock prices (Green and Hwang 2009), financial leverage, liquidity, additions to the S&P 500 index, geographic proximity, and correlated trading:

$$\Delta\mathrm{CORR}_{i,j}^{t_b} = b_{\mathrm{OS}} \cdot \Delta SHM_{i,j}^{t_b} + \lambda_{t_b}$$

$$+b_{\mathrm{MV}_{\mathrm{DIS}}} \cdot \Delta\mathrm{MV}_{\mathrm{DIS}}{}_{i,j}^{t_b} + b_{\mathrm{MV}_{\mathrm{AVG}}} \cdot \Delta\mathrm{MV}_{\mathrm{AVG}}{}_{i,j}^{t_b}$$

$$+b_{P_{\mathrm{DIS}}} \cdot \Delta P_{\mathrm{DIS}}{}_{i,j}^{t_b} + b_{P_{\mathrm{AVG}}} \cdot \Delta P_{\mathrm{AVG}}{}_{i,j}^{t_b}$$

$$+b_{PTBV_{\mathrm{DIS}}} \cdot \Delta PTBV_{\mathrm{DIS}}{}_{i,j}^{t_b} + b_{PTBV_{\mathrm{AVG}}} \cdot \Delta PTBV_{\mathrm{AVG}}{}_{i,j}^{t_b}$$

$$+b_{\mathrm{PTBV}_{\mathrm{DIS}}} \cdot \Delta\mathrm{PTBV}_{\mathrm{DIS}}{}_{i,j}^{t_b} + b_{\mathrm{PTBV}_{\mathrm{AVG}}} \cdot \Delta\mathrm{PTBV}_{\mathrm{AVG}}{}_{i,j}^{t_b}$$

$$+b_{LR_{\mathrm{DIS}}} \cdot \Delta\mathrm{LR}_{\mathrm{DIS}}{}_{i,j}^{t_b} + b_{LR_{\mathrm{AVG}}} \cdot \Delta\mathrm{LR}_{\mathrm{AVG}}{}_{i,j}^{t_b}$$

$$+b_{\mathrm{TA}_{\mathrm{DIS}}} \cdot \Delta\mathrm{TA}_{\mathrm{DIS}}{}_{i,j}^{t_b} + b_{\mathrm{TA}_{\mathrm{AVG}}} \cdot \Delta\mathrm{TA}_{\mathrm{AVG}}{}_{i,j}^{t_b}$$

$$+b_{\mathrm{VO}_{\mathrm{DIS}}} \cdot \Delta\mathrm{VO}_{\mathrm{DIS}}{}_{i,j}^{t_b} + b_{\mathrm{VO}_{\mathrm{AVG}}} \cdot \Delta\mathrm{VO}_{\mathrm{AVG}}{}_{i,j}^{t_b}$$

$$+b_{LOC} \cdot SAME_{LOC}{}_{i,j}^{t_b}$$

Table 3 Correlation matrix

	1	2	3	4	5	6	7	8	9	10	11
1	1.00	0.65	0.87	−0.02	0.24	−0.11	0.23	−0.04	0.14	−0.01	0.07
2	0.65	1.00	0.75	−0.01	0.15	−0.05	0.15	−0.03	0.10	−0.00	0.05
3	0.87	0.75	1.00	−0.02	0.21	−0.11	0.21	−0.04	0.12	−0.01	0.05
4	−0.02	−0.01	−0.02	1.00	−0.02	0.22	−0.04	0.08	−0.02	0.04	−0.02
5	0.24	0.15	0.21	−0.02	1.00	−0.14	0.78	−0.05	0.46	−0.02	0.25
6	−0.11	−0.05	−0.11	0.22	−0.14	1.00	−0.22	0.05	−0.07	0.04	−0.05
7	0.23	0.15	0.21	−0.04	0.78	−0.22	1.00	−0.05	0.39	−0.01	0.25
8	−0.04	−0.03	−0.04	0.08	−0.05	0.05	−0.05	1.00	−0.22	0.04	−0.03
9	0.14	0.10	0.12	−0.02	0.46	−0.07	0.39	−0.22	1.00	−0.01	0.12
10	−0.01	−0.00	−0.01	0.04	−0.02	0.04	−0.01	0.04	−0.01	1.00	−0.19
11	0.07	0.05	0.05	−0.02	0.25	−0.05	0.25	−0.03	0.12	−0.19	1.00
12	−0.01	−0.01	−0.00	0.01	−0.02	−0.00	−0.02	0.06	0.07	0.01	−0.03
13	−0.02	−0.00	−0.01	−0.01	−0.02	−0.00	−0.04	−0.13	0.55	0.01	−0.09
14	−0.03	−0.02	−0.03	0.02	−0.01	0.16	−0.02	0.03	−0.01	−0.01	−0.01
15	0.08	0.04	0.07	−0.02	0.16	−0.04	0.26	0.01	−0.14	0.01	−0.01
16	−0.03	−0.01	−0.03	0.04	−0.04	0.01	−0.02	0.02	−0.03	0.01	−0.02
17	0.10	0.08	0.09	0.01	−0.10	−0.01	−0.04	0.01	−0.07	0.01	−0.04
18	−0.02	−0.03	−0.03	0.01	−0.08	0.03	−0.08	0.02	−0.07	0.01	−0.03
19	0.04	0.03	0.04	−0.00	0.04	−0.02	0.05	0.00	0.01	−0.01	0.02
20	0.00	0.00	0.01	0.00	0.01	−0.01	0.02	−0.00	−0.00	−0.00	0.01
21	0.21	0.11	0.21	0.00	0.05	−0.05	0.12	0.00	−0.01	−0.00	0.00

	12	13	14	15	16	17	18	19	20	21
1	−0.01	−0.02	−0.03	0.08	−0.03	0.10	−0.02	0.04	0.00	0.21
2	−0.01	−0.00	−0.02	0.04	−0.01	0.08	−0.03	0.03	0.00	0.11
3	−0.00	−0.01	−0.03	0.07	−0.03	0.09	−0.03	0.04	0.01	0.21
4	0.01	−0.01	0.02	−0.02	0.04	0.01	0.01	−0.00	0.00	0.00
5	−0.02	−0.02	−0.01	0.16	−0.04	−0.10	−0.08	0.04	0.01	0.05
6	−0.00	−0.00	0.16	−0.04	0.01	−0.01	0.03	−0.02	−0.01	−0.05
7	−0.02	−0.04	−0.02	0.26	−0.02	−0.04	−0.08	0.05	0.02	0.12
8	0.06	−0.13	0.03	0.01	0.02	0.01	0.02	0.00	−0.00	0.00
9	0.07	0.55	−0.01	−0.14	−0.03	−0.07	−0.07	0.01	−0.00	−0.01
10	0.01	0.01	−0.01	0.01	0.01	0.01	0.01	−0.01	−0.00	−0.00
11	−0.03	−0.09	−0.01	−0.01	−0.02	−0.04	−0.03	0.02	0.01	0.00
12	1.00	0.13	0.02	−0.00	−0.01	−0.00	−0.00	−0.00	0.01	0.00
13	0.13	1.00	−0.03	0.10	−0.01	0.00	−0.02	−0.02	0.01	−0.01
14	0.02	−0.03	1.00	−0.06	0.06	−0.01	−0.00	−0.01	−0.00	−0.02
15	−0.00	0.10	−0.06	1.00	−0.01	0.11	−0.02	0.03	0.03	0.09
16	−0.01	−0.01	0.06	−0.01	1.00	−0.04	0.01	−0.01	0.00	−0.02
17	−0.00	0.00	−0.01	0.11	−0.04	1.00	−0.00	−0.03	0.00	0.04
18	−0.00	−0.02	−0.00	−0.02	0.01	−0.00	1.00	0.01	−0.00	−0.01
19	−0.00	−0.02	−0.01	0.03	−0.01	−0.03	0.01	1.00	0.00	0.05
20	0.01	0.01	−0.00	0.03	0.00	0.00	−0.00	0.00	1.00	0.01
21	0.00	−0.01	−0.02	0.09	−0.02	0.04	−0.01	0.05	0.01	1.00

Table 3 (continued)

This table presents the Pearson correlation coefficients all relevant independent variables. Variables are represented by the numbers that are assigned to them in Table 1

$$+b_{add} \cdot ADD_{i,j}^{t_b-1,t_b}$$

$$+b_{\mathrm{CORR_{VO}}} \cdot \Delta \mathrm{CORR}_{VO_{i,j}}{}^{t_b}$$

$$+b_{\Delta HHI} \cdot \Delta HHI_{Ind_{i,j}}^{t_b} \tag{34}$$

These control variables mitigate potential confounding factors by ensuring that our coefficient for changes in common ownership is not influenced by other underlying factors. Specifically, by including common risk factors from Fama and French (2015), we control for market-wide risks that could also be driving changes in common ownership. Additionally, by including variables related to stock prices, financial leverage, and liquidity, we account for financial dynamics that could influence both common ownership and the correlations we observe.

Furthermore, we incorporate variables for S&P 500 index additions and correlated trading to rule out the possibility that external events, such as the inclusion of companies to the index, are driving our results rather than common ownership itself. Finally, by controlling for changes in industry competition with the HHI, we ensure that the effects we observe are not simply due to shifts in market concentration within industries. Overall, our approach can be summarized as follows (see Fig. 2):

1. Create company pairs within each industry super-sector.
2. Compute pairwise correlations and three measures of common ownership for each quarter.
3. Compute the average level and the similarity, i.e., difference in quantiles, of control variables, respectively.
4. Compute changes over one year in pairwise correlation, common ownership, and the control variables.

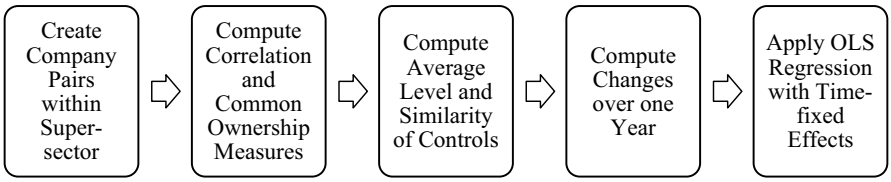

Fig. 2 Flowchart of methodology. This chart illustrates the key steps of our approach. First, we create company pairs within each super-sector. Second, pairwise correlations and common ownership measures are calculated. Third, we compute the average level and the similarity, i.e., the difference in quantiles, of control variables for each company pair. Fourth, we compute changes over one year that are used in OLS regressions with time fixed effects. In the final step, we regress changes in pairwise correlation on the previous year's changes in common ownership and control variables

5. Regress changes in pairwise correlation over previous year's changes in common ownership and control variables, with the inclusion of time fixed effects.

Because the data set covers companies from various industry classifications, it is likely that the data contains heterogeneous patterns across the different business sectors. To account for these, in additional regressions, we divide the dataset into subsamples for each super-sector and estimate the slopes from Eq. (34) for each subset of data, i.e., for each super-sector. To avoid multicollinearity, changes in industry, HHIs are excluded from these regressions.

As an extension of our analysis, we distinguish between short-term and non-short-term common ownership. The underlying idea is that investors with longer investment horizons have stronger incentives and greater ability to influence competition strategically, while short-term investors lack both the incentive and capacity to do so. Based on the proposed competition channel, we expect the positive relationship between common ownership and pairwise correlation to be primarily driven by non-short-term investors. To implement this, we classify each investor as either short-term or non-short-term for every stock and boundary date in our sample, using a symmetric two-year window around that date.[9] An investor is classified as non-short-term for a stock and boundary date if their ownership share for that stock is greater than zero throughout the entire window. Based on this classification, we create two subsamples: one including only holdings of non-short-term investors and the other including only holdings of short-term investors for each stock and boundary date. We then compute the ownership measures for each subsample according to Eqs. (2), (4), and (5) and include them separately in the regressions.

All regression coefficients are estimated using the ordinary least squares (OLS) method.

4 Results

This section presents the empirical results. Table 4 shows the estimations obtained from the univariate regressions based on the pooled data according to Eq. (32). For comparison, Table 4 also depicts the results based on period-wise estimation and subsequent averaging of the estimated coefficients as suggested by Fama and MacBeth (1973). For the period-wise regression, standard errors are calculated according to Newey and West (1987).[10] In both cases, all three measures of common ownership yield positive coefficients with t-values considerably larger than the proposed value of 3.0 by Harvey et al. (2016). This confirms the hypothesized positive relationship between common ownership and pairwise stock return correlation. Note that the threshold of 3.0 is specific to the asset pricing literature due to the unique characteristics of financial data. We chose to adhere to this stricter threshold because

[9] Note that our results for non-short-term common ownership remain positive and statistically significant even when varying the length of the time window used for classification.

[10] For the calculation of Newey and West (1987) standard errors, a lag of $L = 2$ is used.

Table 4 Results of univariate regressions. Dependent variable: Change in the pairwise correlation of stock returns

	Measure	Intercept	Common ownership
Cross-period	ΔSHO	012*** (26.731)	0.157*** (16.261)
	ΔSHC	0.01*** (32.884)	0.015*** (3.544)
	ΔSHB	0.013*** (30.329)	0.072*** (8.844)
Period-wise	ΔSHO	0.014 (0.944)	0.152*** (3.234)
	ΔSHC	0.015 (1.043)	0.055*** (4.093)
	ΔSHB	0.014 (0.977)	0.134*** (3.594)

The presented values are the estimates; t-values are depicted in parenthesis. ΔSHO, ΔSHC, and ΔSHB are the three measures of the change of common ownership. The period under consideration ranges from 1998 to 2022. Coefficients in the "Cross-Period" section are estimated using the full sample and OLS. In the "Period-Wise" section, coefficients are estimated for each period separately using OLS and Newey–West standard errors. The reported coefficients are the averages from all regression results. *T*-values and *p*-values are computed using a two-sided *t* test. $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$

our study deals with financial data, and our analysis of pairwise correlation of stock returns is related to the field of asset pricing.

Table 5 presents the results when incorporating control variables. The coefficients for all three measures of common ownership remain unchanged regarding their direction and statistically significant at the 1%-level. ΔSHO yields a regression slope of 0.08 with a t-value of 10.808.

According to this, an increase in the change in common ownership of one standard deviation (0.044) increases pairwise correlations by $0.08 \cdot 0.044 = 0.352$ percentage points. The coefficients of the vector-based measurements ΔSHC and ΔSHB imply that a one standard deviation increase in these measures leads to an increase in pairwise correlations of $0.032 \cdot 0.102 = 0.3264$ percentage points or $0.090 \cdot 0.051 = 0.4590$ percentage points, respectively. This is in line with our hypothesis that common ownership leads to anticompetitive effects that drive pairwise correlations.

Regarding the slopes of the control variables, the following can be observed. ΔMV_{AVG} yields a statistically significant, positive coefficient. Thus, higher market values are associated with an increase in pairwise correlations. This is consistent with the results of Green and Hwang (2009), and the intuition that larger firms operate in several business areas and are therefore more likely to have overlapping activities with other companies. However, the similarity of company size does not appear to have a substantial influence (ΔMV_{DIS}), suggesting that more similar companies in terms of size do not have a stronger return correlation.

The price difference ΔP_{DIS} has a significant and negative coefficient for all three measures of common ownership. This provides support for the findings of Green and Hwang (2009), as a decrease in the price difference ($\Delta PD < 0$), i.e., more similar prices, is accompanied by a subsequent increase in correlation. Additionally, we

Table 5 Regression results with time fixed effects and control variables

	ΔSHO	ΔSHC	ΔSHB
Common ownership	0.080*** (10.891)	0.032*** (10.511)	0.090*** (14.664)
ΔMV_{DIS}	−0.000 (−0.124)	−0.002 (−0.628)	0.001 (0.148)
ΔMV_{AVG}	0.056*** (6.197)	0.055*** (6.029)	0.054*** (5.879)
ΔP_{DIS}	−0.008*** (−2.963)	−0.008*** (−3.015)	−0.008*** (−2.977)
ΔP_{AVG}	0.076*** (11.194)	0.080*** (11.953)	0.076*** (11.325)
$\Delta PTBV_{DIS}$	−0.010*** (−4.862)	−0.010*** (−4.928)	−0.009*** (−4.756)
$\Delta PTBV_{AVG}$	−0.015*** (−3.282)	−0.014*** (−3.142)	−0.015*** (−3.333)
$\Delta PTBV_{DIS}$	−0.009*** (−4.417)	−0.009*** (−4.423)	−0.009*** (−4.372)
$\Delta PTBV_{AVG}$	−0.024*** (−7.048)	−0.024*** (−7.026)	−0.023*** (−6.883)
ΔLR_{DIS}	0.003 (1.317)	0.003 (1.304)	0.002 (1.217)
ΔLR_{AVG}	0.020*** (5.096)	0.020*** (4.916)	0.020*** (5.092)
ΔTA_{DIS}	−0.005 (−0.944)	−0.005 (−0.987)	−0.005 (−0.854)
ΔTA_{AVG}	−0.066*** (−5.875)	−0.064*** (−5.720)	−0.066*** (−5.835)
ΔVO_{DIS}	0.002 (1.016)	0.002 (0.801)	0.002 (1.052)
ΔVO_{AVG}	0.026*** (6.571)	0.027*** (6.974)	0.025*** (6.394)
SAME_*LOC*	0.004*** (4.242)	0.004*** (4.227)	0.004*** (4.125)
ADD	0.011*** (8.049)	0.013*** (9.445)	0.010*** (7.520)
$\Delta CORR_{VO}$	0.043*** (45.541)	0.043*** (45.624)	0.043*** (45.561)
ΔHHI_{Ind}	0.039*** (11.676)	0.040*** (11.807)	0.040*** (11.899)

Dependent variable: Change in the pairwise correlation of stock returns. The presented values are the estimates using Eq. (34). ΔSHO, ΔSHC and ΔSHB represent the three measures of the change of common ownership. The row "Common Ownership" refers to the coefficients of the respective measure of common ownership. Control variables are explained in Sect. 2. T-values are presented in parenthesis. $^{*}p<0.05$; $^{**}p<0.01$; $^{***}p<0.001$

find that stocks with higher prices (ΔP_{AVG}) correlate more, which is statistically significant at the 1%-level.

Relatedly, the coefficients for the differences in common factors ($\Delta PTBV_{DIS}$, $\Delta PTBV_{DIS}$, ΔTA_{DIS}) are all negative, and statistically significant for market-to-book value and profitability. This suggests that a higher similarity (i.e., a smaller distance) in common factors is associated with a higher correlation.

Changes in the absolute level of common factors also tend to have a negative fact on comovement, as evidenced by the significant and negative coefficients of $\Delta PTBV_{DIS}$, $\Delta PTBV_{DIS}$, ΔTA_{DIS}. In particular, the negative coefficient of $\Delta PTBV_{AVG}$ indicates that decreases in profit margins are followed by increases in pairwise correlations, which, in turn, is linked to increased competition. This finding is in line with the idea that undercutting profit margins (decreases in $\Delta PTBV_{AVG}$) to gain revenue and market share increases industry competition (Dou et al. 2021) and therefore pairwise correlations. Relatedly, the similarity in profit margin also has a significant and negative effect.

The similarity in terms of leverage $\left(\Delta LR_{DIS}\right)$ does not have a significant effect in our sample. However, the level of leverage $\left(\Delta LR_{AVG}\right)$ is positively related to the correlation in the subsequent period.

While the proxy for liquidity similarity $\left(\Delta VO_{DIS}\right)$ does not have a significant impact on pairwise stock correlations in our sample, we find a statistically significant, positive relationship for the proxy of the average liquidity $\left(\Delta VO_{AVG}\right)$. Thus, we find that an increase in the liquidity level is linked to subsequently higher pairwise stock return correlations, indicating that higher liquidity increases stock return correlations.

Consistent with Li and Zhao (2016), we find that the correlation between stocks from companies with headquarters in the same state is higher, indicating that geographical proximity is linked to comovement.

The variable capturing index additions (*Add*) is significant and positive. This is in line with the results of Barberis et al. (2005) who show that additions to the S&P 500 index are associated with an increase in correlations.

Increases in the correlation of trading volumes are associated with a higher subsequent correlation of stock returns, suggesting that correlated trading activities may be a driver of comovement. This finding is compatible with Lee and Rui (2002) who show that there is a positive relationship between trading volume and return volatility.

The statistically significant and positive coefficient of ΔHHI_{Ind} reflects that an intensifying competition within an industry is accompanied with a subsequent increase in pairwise correlations. This finding is in line with the results of Abdoh and Varela (2017) and demonstrates that our coefficients for common ownership do not merely reflect the effects of industry-level competition.

Table 6 presents the results of the industry-wise analysis.[11] Of the 20 ICB super-sectors, a positive relationship can be observed in 13 super-sectors, of which seven are statistically significant. These are "Technology," "Health Care," "Banks," "Insurance," "Real Estate", "Media," and "Industrial Goods and Services." The "Media" and "Banks" super-sectors have the largest coefficients with values of 0.426 and 0.279, respectively. Only the two super-sectors "Automobiles and Parts" and "Personal Care, Drug and Grocery Stores" have a significantly negative coefficient.

Table 7 presents the average concentration measures and their absolute changes across super-sectors, allowing us to link the coefficients to the level of competition in each industry. It is evident that the two significant negative coefficients are associated with industries having the fewest companies and, consequently, higher levels of concentration. With the exception of Media, significant positive coefficients are found in super-sectors with a larger number of companies and lower to moderate HHIs. Additionally, super-sectors with both the highest and lowest changes in HHIs tend to show negative coefficients, while most super-sectors with positive coefficients exhibit moderate variability in HHIs. However, concentration measures alone

[11] For reasons of presentation, the coefficients of the control variables are not reported. Detailed tables on the individual industries can be requested from the corresponding author.

Table 6 Regression results by ICB super-sectors

ICB super-sector	Number of data points	Common ownership
1010—Technology	23,351	0.142*** (9.058)
1510—Telecommunications	892	0.004 (0.038)
2010—Health Care	26,495	0.155*** (8.768)
3010—Banks	2682	0.279*** (6.526)
3020—Financial Services	4096	− 0.036 (− 0.781)
3030—Insurance	3327	0.154*** (3.136)
3510—Real Estate	8828	0.098*** (3.873)
4010—Automobiles and Parts	226	− 0.417** (− 2.056)
4020—Consumer Products and Services	3831	0.045 (− 1.104)
4030—Media	382	0.426*** (3.369)
4040—Retail	3235	0.005 (0.087)
4050—Travel and Leisure	3670	0.009 (0.206)
4510—Food, Beverage and Tobacco	3398	0.054 (1.040)
4520—Personal Care, Drug and Grocery Stores	1195	− 0.215* (− 1.901)
5010—Construction and Materials	357	− 0.119 (− 0.710)
5020—Industrial Goods and Services	45,626	0.095*** (7.362)
5510—Basic Resources	345	− 0.018 (− 0.088)
5520—Chemicals	583	0.161 (1.390)
6010—Energy	4092	0.015 (0.480)
6510—Utilities	9441	− 0.029 (− 0.913)

Dependent variable: Pairwise correlation of stock returns. For the sake of clarity only the regression estimates for ΔSHO as measure for the change of common ownership are depicted. The column "number of data points" contains the number of data points obtained for the corresponding ICB super-sector and used for the estimation according to Eq. (34)

T-values are presented in parenthesis. *$p < 0.05$; **$p < 0.01$; *** $p < 0.001$

cannot fully explain the observed differences in coefficients, suggesting that further research is needed.

It is worth noting that the super-sectors with the highest numbers of data points all exhibit significant and positive coefficients, while the two significant negative coefficients come from super-sectors with smaller sample sizes. Specifically, "Automobiles and Parts" and "Personal Care, Drug and Grocery Stores" have only 226 and 1195 data points, respectively, whereas the significant positive coefficients for "Industrial Goods and Services," "Health Care," and "Technology" are supported by more than 20,000 data points each. Overall, these results suggest a positive relationship between common ownership and the pairwise correlation of stock returns. This supports our hypothesis that anticompetitive effects of common ownership are present across several industries.

Table 8 presents the regression results when considering different investor holding horizons, distinguishing between short-term and non-short-term common ownership. We find that common ownership by non-short-term investors is associated

Table 7 Concentration measures per industry

ICB Super-Sector	Mean HHI	Mean ΔHHI	Number of companies
1010–Technology	997.43	274.46	69.71
1510–Telecommunications	1979.35	394.14	79.57
2010–Health care	779.84	410.72	80.26
3010–Banks	1228.63	211.47	54.86
3020–Financial services	573.50	195.81	35.81
3030–Insurance	1113.84	765.46	69.73
3510–Real estate	288.41	58.03	14.91
4010–Automobiles and parts	3147.50	1530.95	322.92
4020–Consumer products and services	534.29	109.79	55.95
4030–Media	2576.83	422.18	143.36
4040–Retail	1742.86	559.93	133.81
4050–Travel and leisure	994.40	464.98	77.69
4510–Food, beverage and tobacco	1385.34	517.95	53.40
4520–Personal care, drug and grocery stores	2084.46	338.95	110.25
5010–Construction and materials	662.04	335.55	44.24
5020–Industrial goods and services	672.96	546.36	57.57
5510–Basic resources	1060.83	508.58	146.51
5520–Chemicals	1032.61	240.79	72.13
6010–Energy	1645.72	875.10	113.02
6510–Utilities	405.83	40.61	10.29

This table shows the averages over the period of consideration of the HHI, the absolute change in the HHI, and the number of companies per super-sector

with a statistically significant and positive coefficient on the change in pairwise correlation, while short-term common ownership has either insignificant or negative effects. Given that longer-horizon investors have greater incentives and ability to influence competition strategically, these results provide additional support for the proposed competition channel.

5 Discussion

In this section, we discuss the implications and limitations of our results.

5.1 Implications

Using a current data set and three different measures of common ownership, we find that higher common ownership is associated with higher pairwise correlations. The

Table 8 Regression results when differentiation between short-term and non-short-term investors

	Investment horizon	Common ownership
Cross-Period with controls	Short-Term	−0.047* (−1.649)
	Non-Short-Term	0.050*** (6.688)
Period-Wise	Short-Term	0.109 (0.819)
	Non-Short-Term	0.100*** (2.677)

Dependent variable: Pairwise correlation of stock returns. For the sake of clarity, only the regression estimates for ΔSHO as measure for the change of common ownership are depicted. Coefficients in the "Cross-Period with controls" section are estimated using the full sample, Eq. (34) and OLS. In the "Period-Wise" section, coefficients are estimated according to Eq. (32) for each period separately using OLS and Newey–West standard errors. The reported coefficients are the averages from all regression results. *T*-values and *p*-values are computed using a two-sided *t* test

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

magnitudes of the effects are comparable, although they are smaller when cosine similarity is used. As cosine similarity puts more weight on large investors, this implies that our results are likely to be driven by more than just a few large investors, such as BlackRock or Vanguard. Therefore, our study provides robust evidence for a positive relationship between common ownership at the individual investor level and stock comovement and synchronicity. Furthermore, as competition and pairwise correlation are negatively related according to Abdoh and Varela (2017), our result can be interpreted as support for anticompetitive effects of common ownership. We provide additional support for this competition channel in two ways: first, by including changes in industry-level HHIs as control variables to account for the effects of market concentration, and second, by distinguishing between short-term and non-short-term common ownership. We find that only non-short-term common ownership yields significantly positive regression coefficients, consistent with the idea that longer-horizon investors have stronger incentives and greater ability to influence competition strategically.

When examining effects by industry, we estimate positive coefficients for the majority of super-sectors. Seven of these are statistically significant, including the coefficient in the banking sector. This provides support for the recent results of Iselin et al. (2021) that common ownership increases the comovement of bank stock returns, and the findings of Azar et al. (2022) regarding anticompetitive effects of common ownership in the US banking industry. The positive effect in the "Travel and Leisure" super-sector is not significant. Therefore, we cannot provide additional support for the results regarding the US airline industry reported in Azar et al. (2018). However, our insignificant coefficient is consistent with the reevaluation of Dennis et al. (2022), which concludes that common ownership is likely to not have anticompetitive effects in the airline industry.

The coefficients of our control variables provide current evidence on effects provided in previous research. Furthermore, they allow us to separate the effects of

factor similarity and the absolute level of the respective factor. Inter alia, we can confirm that stocks of larger companies and stocks with similar price levels have higher correlations (Green and Hwang 2009), and the finding of Barberis et al. (2005) that the addition to an index is accompanied by stronger comovement. Furthermore, we find that decreases in profit margins are accompanied by an increase in correlations, supporting the notion that undercutting profit margins increases competition and thus leads to a higher fraction of idiosyncratic risks (Dou et al. 2021).

Overall, our study contributes to the literature on stock return synchronicity and comovement by using current data to demonstrate a robust relationship between common ownership of any investor and investment type, and the pairwise correlation of stock returns. This extends the analysis of Antón and Polk (2014) who focus on actively managed funds. Furthermore, for a large selection of variables discussed in the literature, we show and clarify whether their level or similarity drives comovement. Finally, we link our results to the controversy regarding anticompetitive effects and show that the pairwise correlations of stock returns are consistent with anticompetitive effects of common ownership across a wide range of industries.

5.2 Limitations

Although our data set includes a selection of companies from different sectors and with different market capitalizations, our sample only consists of companies from the USA, and focuses on larger capitalizations. Therefore, the generalizability of our findings to other countries or smaller firms is uncertain. However, given that the institutional investors of our sample are likely to invest in companies across various market sizes and regions (e.g., Döring et al. 2021), we expect the effects observed in our study may also apply to other markets and smaller companies, though this warrants further investigation.

In some super-sectors, the number of companies is small. Therefore, our insignificant coefficients should not be taken as a strong indication of the absence of a positive relationship between common ownership and pairwise correlations. However, most of our significant positive coefficients are in industries with a high number of observations, such as "Technology," "Health Care," and "Industrial Goods and Services," and can therefore be considered a strong indication of the proposed relationship between common ownership and correlation.

Extending the sample to more companies is problematic as the effort of collecting and processing data is substantial, and reports on companies not listed in the S&P 500 are often not available in the data base. However, focusing on the S&P 500 has the advantage of comparable companies, especially regarding the index affiliation, which means that we can rule out that index affiliation drives our results.

Another concern is the possibility of reverse causality, meaning that investors could deliberately choose to invest in companies with higher correlation. Existing research, e.g., Boller and Morton (2020), Barberis et al. (2005) or Antón and Polk (2014), addresses this problem by analyzing external events, such as index additions. In our study, we make use of the time series data and consider how a change in common ownership affects the pairwise correlation in subsequent periods. Our

finding that an increase in common ownership is accompanied by an increase in correlation in later periods indicates that reverse causality is unlikely. Another argument against reverse causality is that portfolio theory suggests that investors with preferences according to the mean–variance framework prefer to invest in stocks with a low correlation, in order to minimize the variance of portfolio returns, and as shown for strategic investors by Münster and Walther (2021). From a portfolio theory perspective, one would therefore expect investors to select stocks that have a low correlation, i.e., common ownership to be associated with a significantly lower correlation. It is therefore unlikely that investors deliberately select highly correlated stocks, which means that reverse causality is unlikely.

Finally, we do not show a direct link between common ownership and competition but rely on the results of Abdoh and Varela (2017) to relate our results on pairwise correlations to competition. While this means our findings do not constitute definitive evidence of anticompetitive effects, they are consistent with such effects across a wide range of industries. Moreover, the inclusion of changes in industry HHIs as control variables and the distinction between short-term and non-short-term common ownership provide additional support to the proposed competition channel. Our study thus contributes both to the literature on stock return comovement and synchronicity, and to the ongoing discussion on the anticompetitive implications of common ownership.

6 Conclusion

Using S&P 500 data from 1998 to 2022, we show that pairwise correlations of stock returns within an industry are positively related with common ownership. Given the negative relationship between competition and correlation, our findings suggest that common ownership may have anticompetitive effects in multiple industries.

Our findings have several policy implications. Antitrust regulations may need to consider indirect competitive effects arising from common ownership from active and passive investors, even if companies do not explicitly coordinate. Our analysis per industry highlights the sectors Technology, Health Care, Banks, Insurance, Real Estate, Media, and Industrial Goods and Services, for which the effects are statistically significant. Enhancing disclosure requirements on ownership structures and improving the availability of data on ownership could help policymakers better monitor common ownership. This would enable researchers to identify anticompetitive effects and investigate potential countermeasures. These include limits on cross-holdings in competing companies within the same industry, implementing policies that reduce voting rights of common owners in competing companies, and enforcing a stricter oversight of board appointments to prevent a disproportionate influence on multiple companies' boards within the same industry. However, when considering limitations on common ownership, policymakers need to consider that this may reduce diversification opportunities, and consequently reduce market stability.

In terms of future research, several important avenues remain. First, while we have addressed reverse causality, additional evidence is needed to confirm whether our proposed mechanism is indeed causal. Second, understanding the heterogeneous effects observed across industries and uncovering the underlying mechanisms present an intriguing research avenue. Third, assessing the generalizability of our findings to different market capitalizations and other countries with different regulatory environments would provide valuable insights. Moreover, examining the long-term implications of common ownership, such as its influence on innovation, is essential for a deeper understanding of its overall welfare impact. Across all these areas, improving the availability and quality of ownership data will be vital to advancing research.

Appendix

Appendix A: results for unadjusted investor names

Appendix A presents the results without adjusting investor names. Table 9 shows the estimated regression coefficients. We can confirm our main results and verify the positive relationship between all three measures of common ownership and the pairwise correlation of stock returns.
All coefficients and t values remain almost unchanged when control variables are included in the regression. This means that our results are robust to not preprocessing investor names (Table 10).

Table 9 Results of univariate regressions based on unadjusted investor names

	Measure	Intercept	Common ownership
Cross-period	ΔSHO	0.012*** (26.989)	0.149*** (15.003)
	ΔSHC	0.014*** (33.141)	0.008** (2.049)
	ΔSHB	0.013*** (30.285)	0.070*** (8.364)
Period-wise	ΔSHO	0.014 (0.939)	0.157*** (3.175)
	ΔSHC	0.015 (1.050)	0.051*** (3.933)
	ΔSHB	0.014 (0.968)	0.138*** (3.767)

Dependent variable: Pairwise correlation of stock returns. The presented values are the estimates; t-values are depicted in parenthesis

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Table 10 Regression results with time fixed effects and control variables based on unadjusted investor names

	ΔSHO	ΔSHC	ΔSHB
Common Ownership	0.081*** (10.587)	0.029*** (9.721)	0.093*** (14.593)
ΔMV_{DIS}	−0.000 (−0.043)	−0.002 (−0.581)	0.001 (0.299)
ΔMV_{AVG}	0.056*** (6.157)	0.055*** (6.016)	0.053*** (5.775)
ΔP_{DIS}	−0.008*** (−2.964)	−0.008*** (−3.031)	−0.008*** (−3.005)
ΔP_{AVG}	0.076*** (11.254)	0.081*** (12.017)	0.077*** (11.394)
$\Delta PTBV_{DIS}$	−0.010*** (−4.852)	−0.010*** (−4.914)	−0.009*** (−4.740)
$\Delta PTBV_{AVG}$	−0.015*** (−3.248)	−0.014*** (−3.036)	−0.015*** (−3.256)
$\Delta PTBV_{DIS}$	−0.009*** (−4.406)	−0.009*** (−4.424)	−0.009*** (−4.349)
$\Delta PTBV_{AVG}$	−0.024*** (−7.005)	−0.023*** (−6.964)	−0.023*** (−6.822)
ΔLR_{DIS}	0.003 (1.336)	0.002 (1.272)	0.002 (1.240)
ΔLR_{AVG}	0.020*** (5.094)	0.020*** (4.872)	0.020*** (5.055)
ΔTA_{DIS}	−0.005 (−0.933)	−0.005 (−1.012)	−0.004 (−0.830)
ΔTA_{AVG}	−0.066*** (−5.883)	−0.064*** (−5.658)	−0.065*** (−5.818)
ΔVO_{DIS}	0.002 (1.019)	0.002 (0.834)	0.002 (1.093)
ΔVO_{AVG}	0.026*** (6.723)	0.028*** (7.108)	0.025*** (6.522)
SAME_*LOC*	0.004*** (4.238)	0.004*** (4.228)	0.004*** (4.120)
ADD	0.011*** (7.879)	0.013*** (9.471)	0.010*** (7.226)
$\Delta CORR_{VO}$	0.043*** (45.546)	0.043*** (45.649)	0.043*** (45.568)
ΔHHI_{Ind}	0.039*** (11.689)	0.040*** (11.769)	0.040*** (11.868)

Dependent variable: Pairwise correlation of stock returns. For reasons of presentation, only particularly relevant control variables are reported. The presented values are the estimates; *t*-values are depicted in parenthesis

*$p<0.05$; **$p<0.01$; ***$p<0.001$

Appendix B: Results for different measures of ownership overlap

Appendix B compares the results for different measures of ownership overlap.

The first measure of ownership overlap $\left(SHO = SHO_{\min}\right)$ is defined according to Eq. (2). The second measure, SHO_{AVG}, drops the penalty term. It is defined as the sum of the average percentage of stocks held by common investors:

$$SHO^{t}_{avg,i,j} = \sum_{k} \frac{S^{t}_{i,k} + S^{t}_{j,k}}{2} \forall i, j, t. \tag{35}$$

In contrast with Antón and Polk (2014), SHO_{AVG} does not take into account the market capitalizations but uses percentages of stocks held.

The third measure SHO_{half} uses half the penalty term and is defined as:

$$SHO^{t}_{half,i,j} = \sum_{k} \left(1 - \frac{1}{2} \cdot \frac{\delta^{t}_{k,i,j}}{S^{t}_{i,k} + S^{t}_{j,k}}\right) \cdot \frac{S^{t}_{i,k} + S^{t}_{j,k}}{2} \forall i, j, t. \tag{36}$$

SHO_{half} corresponds to the average of SHO_{min} and SHO_{AVG}, and therefore represents a compromise between both approaches.

Table 11 provides descriptive statistics for the three measures.

The correlation matrix for the three measurements is depicted in Table 12.

This table presents the Pearson correlation coefficients between the three measures of ownership overlap.

Table 13 presents the results of the plain regression. The direction and significance are the same for all measures considered. Furthermore, the t-values are on a similar level, which indicates that the results are robust to the way of measuring ownership overlap.

Table 11 Descriptive statistics of three alternative measures of ownership overlap

Parameter	Mean	Standard deviation	Min	Max	10th quantile	90th Quantile
SHO_{min}	0.355	0.121	0.000	0.931	0.200	0.494
SHO_{AVG}	0.601	0.165	0.000	2.734	0.413	0.757
SHO_{half}	0.478	0.136	0.000	1.579	0.310	0.621
ΔSHO_{min}	0.016	0.044	−0.375	0.553	−0.029	0.063
ΔSHO_{AVG}	0.018	0.069	−2.215	0.868	−0.041	0.083
ΔSHO_{half}	0.017	0.051	−1.118	0.627	−0.030	0.068

The descriptive statistics for each measure of the ownership overlap are computed with all $n = 167,479$ data points in our sample. The measures of ownership overlap SHO_{min}, SHO_{AVG}, and SHO_{half} are defined according to Eqs. (2), (35) and (36), respectively. Following the procedure described in Eq. (23), the corresponding changes of the variables are computed as the difference between two boundary dates.

Table 12 Correlation matrix for the three alternative measures of ownership overlap

Parameter	ΔSHO_{min}	ΔSHO_{AVG}	ΔSHO_{half}
ΔSHO_{min}	1.00	0.62	0.85
ΔSHO_{AVG}	0.62	1.00	0.94
ΔSHO_{half}	0.85	0.94	1.00

Table 13 Results of univariate regressions considering the three different measures of ownership overlap

	Measure	Intercept	Common ownership
Cross-period	ΔSHO_{min}	0.012*** (26.731)	0.157*** (16.261)
	ΔSHO_{AVG}	0.012*** (27.473)	0.141*** (23.289)
	ΔSHO_{half}	0.011*** (25.481)	0.186*** (22.720)
Period-wise	ΔSHO_{min}	0.014 (0.944)	0.152*** (3.234)
	ΔSHO_{AVG}	0.014 (0.989)	0.084*** (3.431)
	ΔSHO_{half}	0.014 (0.945)	0.125*** (3.442)

Dependent variable: Pairwise correlation of stock returns. The presented values are the estimates; *t*-values are depicted in parenthesis

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Table 14 Regression results with time fixed effects and control variables considering the three different measures of ownership overlap

Measure	Common ownership
ΔSHO_{min}	0.080*** (10.891)
ΔSHO_{AVG}	0.037*** (8.110)
ΔSHO_{half}	0.064*** (10.213)

Dependent variable: Pairwise correlation of stock returns. For reasons of presentation, only particularly relevant control variables are reported. The presented values are the estimates; t-values are depicted in parenthesis

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Table 14 presents the results when including control variables. The signs and significance of all coefficients remain qualitatively unchanged. This indicates that our main results are not driven by our choice of measuring ownership overlap.[12]

Appendix C: results for ownership vectors with an alternative size of 15,000 and 1500 dimensions

In Appendix C, the results for different dimensions of the ownership vectors are presented. As depicted in Table 15, the results of the univariate regression remain qualitatively unchanged when an ownership vector with dimension $K = 15{,}000$ (instead of $K = 5000$) is used in the analysis. All three measures of common ownership yield positive and statistically significant coefficients.
The result is robust to the inclusion of control variables and time fixed effects. Table 16 presents the corresponding regression coefficients.

Table 15 Results of univariate regressions for an ownership vector with dimension $K = 15{,}000$

	Measure	Intercept	Common ownership
Cross-period	ΔSHO	0.012*** (26.731)	0.157*** (16.261)
	ΔSHC	0.014*** (32.735)	0.017*** (4.085)
	ΔSHB	0.013*** (30.179)	0.078*** (9.269)
Period-wise	ΔSHO	0.014 (0.944)	0.152*** (3.234)
	ΔSHC	0.015 (1.052)	0.051*** (4.114)
	ΔSHB	0.014 (0.986)	0.131*** (3.348)

Dependent variable: Pairwise correlation of stock returns. The presented values are the estimates; *t*-values are depicted in parenthesis

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

[12] For reasons of presentation, the coefficients of the control variables are not reported. Detailed tables on the individual industries can be provided by the corresponding author.

Table 16 Regression results with time fixed effects and control variables for an ownership vector with dimension $K = 15{,}000$

	ΔSHO	ΔSHC	ΔSHB
Common Ownership	0.080*** (10.891)	0.028*** (9.192)	0.083*** (12.910)
ΔMV_{DIS}	−0.000 (−0.124)	−0.002 (−0.587)	0.000 (0.096)
ΔMV_{AVG}	0.056*** (6.197)	0.055*** (6.034)	0.054*** (5.896)
ΔP_{DIS}	−0.008*** (−2.963)	−0.008*** (−3.038)	−0.008*** (−2.979)
ΔP_{AVG}	0.076*** (11.194)	0.081*** (12.012)	0.076*** (11.342)
$\Delta PTBV_{DIS}$	−0.010*** (−4.862)	−0.010*** (−4.965)	−0.010*** (−4.808)
$\Delta PTBV_{AVG}$	−0.015*** (−3.282)	−0.014*** (−3.116)	−0.015*** (−3.346)
$\Delta PTBV_{DIS}$	−0.009*** (−4.417)	−0.009*** (−4.452)	−0.009*** (−4.393)
$\Delta PTBV_{AVG}$	−0.024*** (−7.048)	−0.023*** (−6.961)	−0.023*** (−6.890)
ΔLR_{DIS}	0.003 (1.317)	0.002 (1.271)	0.002 (1.208)
ΔLR_{AVG}	0.020*** (5.096)	0.020*** (4.876)	0.020*** (5.101)
ΔTA_{DIS}	−0.005 (−0.944)	−0.005 (−1.011)	−0.005 (−0.883)
ΔTA_{AVG}	−0.066*** (−5.875)	−0.065*** (−5.784)	−0.066*** (−5.870)
ΔVO_{DIS}	0.002 (1.016)	0.002 (0.784)	0.002 (0.996)
ΔVO_{AVG}	0.026*** (6.571)	0.028*** (7.180)	0.026*** (6.687)
SAME_*LOC*	0.004*** (4.242)	0.004*** (4.215)	0.004*** (4.135)
ADD	0.011*** (8.049)	0.013*** (9.567)	0.011*** (7.831)
$\Delta CORR_{VO}$	0.043*** (45.541)	0.043*** (45.617)	0.043*** (45.572)
ΔHHI_{Ind}	0.039*** (11.676)	0.040*** (11.726)	0.040*** (11.798)

Dependent variable: Pairwise correlation of stock returns. For reasons of presentation, only particularly relevant control variables are reported. The presented values are the estimates; *t*-values are depicted in parenthesis.

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

In addition to the previous results, Tables 17 and 18 show the empirical results when a vector size of $K = 1500$ is used for the ownership vectors. Again, all coefficients yield statically significant and positive coefficients in all constellations.

Table 17 Regression results with time fixed effects and control variables for an ownership vector with dimension $K = 1500$

	Measure	Intercept	Common ownership
Cross-period	ΔSHO	0.012*** (26.731)	0.157*** (16.261)
	ΔSHC	0.014*** (32.735)	0.020*** (5.047)
	ΔSHB	0.013*** (30.717)	0.064*** (8.544)
Period-wise	ΔSHO	0.014 (0.944)	0.152*** (3.234)
	ΔSHC	0.015 (1.046)	0.054*** (3.936)
	ΔSHB	0.014 (0.976)	0.129*** (3.517)

Dependent variable: Pairwise correlation of stock returns. The presented values are the estimates; t-values are depicted in parenthesis

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Table 18 Regression results with time fixed effects and control variables for an ownership vector with dimension $K = 1{,}500$

	ΔSHO	ΔSHC	ΔSHB
Common ownership	0.080*** (10.891)	0.034*** (12.070)	0.094*** (16.918)
ΔMV_{DIS}	−0.000 (−0.124)	−0.002 (−0.631)	0.001 (0.202)
ΔMV_{AVG}	0.056*** (6.197)	0.056*** (6.127)	0.055*** (6.024)
ΔP_{DIS}	−0.008*** (−2.963)	−0.008*** (−3.026)	−0.008*** (−2.952)
ΔP_{AVG}	0.076*** (11.194)	0.081*** (11.992)	0.076*** (11.272)
$\Delta PTBV_{DIS}$	−0.010*** (−4.862)	−0.010*** (−4.950)	−0.009*** (−4.747)
$\Delta PTBV_{AVG}$	−0.015*** (−3.282)	−0.015*** (−3.272)	−0.016*** (−3.444)
$\Delta PTBV_{DIS}$	−0.009*** (−4.417)	−0.009*** (−4.417)	−0.009*** (−4.372)
$\Delta PTBV_{AVG}$	−0.024*** (−7.048)	−0.024*** (−6.976)	−0.023*** (−6.893)
ΔLR_{DIS}	0.003 (1.317)	0.002 (1.206)	0.002 (1.161)
ΔLR_{AVG}	0.020*** (5.096)	0.020*** (4.965)	0.021*** (5.149)
ΔTA_{DIS}	−0.005 (−0.944)	−0.005 (−1.002)	−0.005 (−0.855)
ΔTA_{AVG}	−0.066*** (−5.875)	−0.064*** (−5.720)	−0.067*** (−5.938)
ΔVO_{DIS}	0.002 (1.016)	0.002 (0.808)	0.003 (1.122)
ΔVO_{AVG}	0.026*** (6.571)	0.027*** (6.868)	0.024*** (6.080)
SAME_*LOC*	0.004*** (4.242)	0.004*** (4.218)	0.004*** (4.115)
ADD	0.011*** (8.049)	0.012*** (9.369)	0.010*** (7.237)
$\Delta CORR_{VO}$	0.043*** (45.541)	0.043*** (45.600)	0.043*** (45.492)
ΔHHI_{Ind}	0.039*** (11.676)	0.040*** (11.761)	0.040*** (11.920)

Dependent variable: Pairwise correlation of stock returns. The presented values are the estimates; t-values are depicted in parenthesis

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

Appendix D: results when using an alternative time window of two years length

Appendix D summarizes the empirical analysis when a subsequent time window of two years length (instead of one year) is used for the comparison and estimation of the correlation coefficient. As illustrated in Table 19, all three measures of

Table 19 Results of univariate regressions considering a subsequent time window of two years length for the correlation estimation

	Measure	Intercept	Common ownership
Cross-period	ΔSHO	0.018*** (25.981)	0.134*** (12.261)
	ΔSHC	0.021*** (31.667)	0.036*** (7.480)
	ΔSHB	0.019*** (27.648)	0.119*** (12.388)
Period-wise	ΔSHO	0.022 (0.990)	0.092* (1.953)
	ΔSHC	0.025 (1.078)	0.025** (1.987)
	ΔSHB	0.023 (0.989)	0.102*** (3.647)

Dependent variable: Pairwise correlation of stock returns. The presented values are the estimates; t-values are depicted in parenthesis

*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$

common ownership have a positive impact on pairwise stock return correlations. Additionally, all coefficients when following the procedure described by Fama and MacBeth (1973) are positive and statistically significant.

Table 20 presents the results when controlling for time fixed effects and all the control variables. In line with the results presented in Table 6, all three coefficients for common ownership have the expected sign. The values of the coefficients are

Table 20 Regression results with time fixed effects and control variables considering a subsequent time window of two years length for the correlation estimation

	ΔSHO	ΔSHC	ΔSHB
Common ownership	0.053*** (6.632)	0.019*** (5.682)	0.086*** (12.437)
ΔMV_{DIS}	0.000 (0.139)	−0.001 (−0.230)	0.002 (0.590)
ΔMV_{AVG}	−0.024** (−2.555)	−0.022** (−2.404)	−0.026*** (−2.785)
ΔP_{DIS}	−0.013*** (−4.762)	−0.013*** (−4.877)	−0.013*** (−4.815)
ΔP_{AVG}	0.047*** (6.959)	0.050*** (7.348)	0.046*** (6.742)
$\Delta PTBV_{DIS}$	−0.014*** (−6.333)	−0.014*** (−6.399)	−0.014*** (−6.176)
$\Delta PTBV_{AVG}$	0.004 (0.876)	0.005 (0.998)	0.003 (0.560)
$\Delta PTBV_{DIS}$	−0.014*** (−6.166)	−0.014*** (−6.156)	−0.014*** (−6.167)
$\Delta PTBV_{AVG}$	−0.013*** (−3.762)	−0.013*** (−3.800)	−0.013*** (−3.732)
ΔLR_{DIS}	0.005** (2.530)	0.005** (2.474)	0.005** (2.430)
ΔLR_{AVG}	−0.004 (−1.007)	−0.005 (−1.153)	−0.003 (−0.763)
ΔTA_{DIS}	−0.008 (−1.483)	−0.009 (−1.595)	−0.007 (−1.387)
ΔTA_{AVG}	−0.044*** (−3.835)	−0.044*** (−3.800)	−0.048*** (−4.145)
ΔVO_{DIS}	−0.013*** (−4.534)	−0.014*** (−4.679)	−0.013*** (−4.329)
ΔVO_{AVG}	−0.036*** (−7.277)	−0.034*** (−6.862)	−0.038*** (−7.640)
SAME_*LOC*	0.008*** (5.983)	0.008*** (5.968)	0.008*** (5.824)
ADD	0.011*** (8.309)	0.013*** (9.365)	0.010*** (7.391)
$\Delta CORR_{VO}$	0.059*** (42.702)	0.060*** (42.817)	0.059*** (42.698)
ΔHHI_{Ind}	0.042*** (13.997)	0.043*** (14.160)	0.043*** (14.245)

Dependent variable: Pairwise correlation of stock returns. For reasons of presentation, only particularly relevant control variables are reported. The presented values are the estimates; *t*-values are depicted in parenthesis

*$p<0.05$; **$p<0.01$; ***$p<0.001$

Table 21 Results of univariate regressions considering a surrounding time window for the correlation estimation

	Measure	Intercept	Common ownership
Cross-period	ΔSHO	0.014*** (29.974)	0.219*** (21.047)
	ΔSHC	0.016*** (37.423)	0.027*** (6.236)
	ΔSHB	0.015*** (33.271)	0.143*** (15.995)
Period-wise	ΔSHO	0.016 (0.934)	0.208*** (3.705)
	ΔSHC	0.018 (1.062)	0.064*** (4.057)
	ΔSHB	0.016 (0.974)	0.201*** (4.794)

Dependent variable: Pairwise correlation of stock returns. The presented values are the estimates; *t*-values are depicted in parenthesis

*$p<0.05$; **$p<0.01$; ***$p<0.001$

Table 22 Regression results with time fixed effects and control variables considering a surrounding time window for the correlation estimation

	ΔSHO	ΔSHC	ΔSHB
Common Ownership	0.159*** (20.735)	0.050*** (16.487)	0.168*** (25.988)
ΔMV_{DIS}	0.017*** (4.920)	0.013*** (3.906)	0.018*** (5.364)
ΔMV_{AVG}	−0.048*** (−5.348)	−0.049*** (−5.464)	−0.054*** (−5.951)
ΔP_{DIS}	−0.005* (−1.912)	−0.005** (−2.048)	−0.005* (−1.916)
ΔP_{AVG}	0.056*** (8.362)	0.066*** (9.980)	0.058*** (8.779)
$\Delta PTBV_{DIS}$	−0.022*** (−11.297)	−0.022*** (−11.461)	−0.022*** (−11.141)
$\Delta PTBV_{AVG}$	0.016*** (3.581)	0.018*** (4.035)	0.016*** (3.510)
$\Delta PTBV_{DIS}$	−0.008*** (−4.094)	−0.008*** (−4.091)	−0.008*** (−4.040)
$\Delta PTBV_{AVG}$	0.017*** (4.986)	0.017*** (5.064)	0.018*** (5.405)
ΔLR_{DIS}	0.003* (1.672)	0.003 (1.595)	0.003 (1.483)
ΔLR_{AVG}	0.019*** (4.760)	0.017*** (4.283)	0.019*** (4.725)
ΔTA_{DIS}	0.005 (0.885)	0.004 (0.770)	0.005 (1.024)
ΔTA_{AVG}	−0.015 (−1.335)	−0.011 (−0.957)	−0.013 (−1.200)
ΔVO_{DIS}	−0.004* (−1.789)	−0.005** (−2.098)	−0.004* (−1.818)
ΔVO_{AVG}	−0.030*** (−7.767)	−0.026*** (−6.854)	−0.030*** (−7.817)
SAME_*LOC*	0.004*** (4.408)	0.004*** (4.401)	0.004*** (4.214)
ADD	0.007*** (5.219)	0.010*** (7.923)	0.006*** (4.616)
$\Delta CORR_{VO}$	0.051*** (55.130)	0.051*** (55.423)	0.051*** (55.036)
ΔHHI_{Ind}	0.026*** (7.779)	0.026*** (8.010)	0.027*** (8.138)

Dependent variable: Pairwise correlation of stock returns. For reasons of presentation, only particularly relevant control variables are reported. The presented values are the estimates; *t*-values are depicted in parenthesis

*$p<0.05$; **$p<0.01$; ***$p<0.001$

smaller in the case of a time window with a length of two years. However, the coefficients of all three measures of common ownership are positive and statistically significant at the 1%-level.

Appendix E: results when using surrounding time windows

Appendix E presents the empirical results when considering a surrounding time window of one year, instead of the subsequent year, for the estimation of the correlation coefficients as described by Eq. (29). Table 21 shows the results for the univariate regressions that confirm the positive relationship between common ownership and pairwise stock return correlations. The coefficients for all three measurements of common ownership are higher than their counterparts in the main article.
Table 22 summarizes the results when considering all control variables. In this constellation, all three measures of common ownership have a positive and statistically significant coefficient that is higher than in the case of a subsequent time window (see Table 5).

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Author contribution Both authors contributed significantly to conceptualization, methodology, data collection, and the discussion. M.M. implemented the analysis and prepared the tables. M.W. wrote the initial draft of the manuscript. Both authors reviewed and edited the manuscript.

Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

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